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Doubly-excited $^{1,3}\text{D}^e$ resonance states of two-electron positive ions Li^+ and Be^{2+} in Debye plasmas

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We investigate the bound $^{1,3}\text{D}$ states and the doubly-excited $^{1,3}\text{D}^e$ resonance states of two-electron positive ions Li^+ and Be^{2+} by employing correlated exponential wave functions. In the framework of the stabilization method, we are able to extract three series ($2pnp$, $2snd$, $2pnf$) of $^{1,3}\text{D}^e$ resonances and two series ($2pnp$, $2snd$) of $^{3,1}\text{D}^e$ resonances below the $N=2$ threshold. The $^{1,3}\text{D}^e$ resonance parameters (resonance energies and widths) for Li^+ and Be^{2+} along with the bound-excited $1s3d$ $^{1,3}\text{D}$ state energies are reported for the first time as functions of the screening parameter. Accurate resonance energies and widths are also reported for Li^+ and Be^{2+} in vacuum. For free-atomic cases, comparisons are made with the reported results and few resonance states are reported for the first time. © 2014 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4862036>]

I. INTRODUCTION

The study of atomic resonances in plasma environment has gained tremendous attention in the recent years^{1–15} and is one of the interesting topics of current research.^{9–15} Recently, resonance states of the helium atom and the hydrogen negative ion in Debye plasmas have been studied in the framework of the stabilization method^{14,15} and the complex-coordinate rotation method using the Hylleraas-type wave functions.^{9–12} Investigations on resonance states of helium in Debye plasmas have also been reported by Ordóñez-Lasso *et al.*¹³ using the configuration interaction method based on the explicitly correlated configurations. There have been a number of studies using the stabilization method to probe the resonance states of the positronium negative ion¹⁶ and the exotic molecular ions¹⁷ in plasma environments. It should be mentioned here that Winkler and coworkers¹⁸ have shown the existence of low-lying resonances in the s-wave channel of screened Coulomb potentials and pointed out likely importance of such phenomena for resonant radiative recombination processes in sufficiently dense plasmas. Applications of atomic processes in plasma environments can be found in detail from earlier articles.^{19–39} Recently, Chang *et al.*³⁹ reported an interesting study related to photoionization of atoms in Debye plasmas.

In the present work, we have calculated the bound $^{1,3}\text{D}$ states and the doubly-excited $^{1,3}\text{D}^e$ resonance states of two-electron positive ions Li^+ and Be^{2+} below the $N=2$ thresholds of the respective subsystems. In the free-atomic cases, $^{1,3}\text{D}^e$ resonance states of Li^+ and B^+ ion have been reported by Bhatia and Ho.⁴⁰ In this work, we have employed the stabilization method^{1,2,14,22,23,41–45} to extract

resonance parameters. Highly correlated exponential wave functions are used to describe the two-electron positive ions Li^+ and Be^{2+} . The convergence of the calculations has been examined with increasing number of terms in the basis functions. Atomic units (a.u.) are used throughout this work.

II. HAMILTONIAN AND WAVE FUNCTIONS

The non-relativistic screened Hamiltonian H for the proposed systems can be written as

$$H = -\frac{1}{2}[\nabla_1^2 + \nabla_2^2] - Z[V(r_1) + V(r_2)] + V(r_{12}), \quad (1)$$

$$V(r) = \frac{1}{r} \exp(-\mu r), \quad (2)$$

where r_1 and r_2 are the radial coordinates of the two electrons and r_{12} is their relative distance. A particular value of the screening parameter μ corresponds to the range of plasma conditions, as the Debye parameter is a function of electron density and electron temperature. $\mu=0$ indicates the unscreened case. For the two-component plasma near thermodynamic equilibrium, the Debye screening length μ takes the form^{27–29}

$$\mu = \sqrt{\frac{4\pi(1+Z^*)n}{k_B T}}, \quad (3)$$

where k_B is the Boltzmann constant, n and T are the number density of the plasma electrons and temperature, respectively, and Z^* is the effective charge of the ions in the embedded plasma.

For $^{1,3}\text{D}^e$ states, we employ highly correlated wave functions

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TABLE I. The bound 1^3D states of Li^+ and Be^{2+} for different values of the Debye screening parameter, along with the $1s^2S$ threshold energies of Li^{2+} and Be^{3+} .

$1/\mu$	$\text{Li}^+(1s3d^1D)$	$\text{Li}^+(1s3d^3D)$	$\text{Li}^{2+}(1s^2S)$
∞	-4.72239098976 -4.72239098974 ^a -4.7223909884 ^b	-4.72252691378 -4.72252691377 ^a -4.7225269124 ^b	-4.5000000000
200	-4.69753920783	-4.69767498640	-4.4850187292
100	-4.67297894144	-4.67311429366	-4.4700748341
50	-4.62470604330	-4.62483975690	-4.4402986787
20	-4.48627645047	-4.48639985130	-4.3518546295
10	-4.27504959753 -4.27504959752 ^a	-4.27514278904 -4.27514278903 ^a	-4.2073405204
	$\text{Be}^{2+}(1s3d^1D)$	$\text{Be}^{2+}(1s3d^3D)$	$\text{Be}^{3+}(1s^2S)$
∞	-8.50021582809 -8.50021582807 ^a -8.5002158256 ^b	-8.50058234443 -8.50058234442 ^a -8.5005823430 ^b	-8.0000000000
200	-8.46536445927	-8.46573093586	-7.9800187344
100	-8.43080751393	-8.43117334559	-7.9600748754
50	-8.36255682755	-8.36292066782	-7.9202990068
20	-8.16441742778	-8.16476829879	-7.8018596373
10	-7.85470283142 -7.85470283139 ^a	-7.85501307593 -7.85501307592 ^a	-7.6073791029
5	-7.3052849491 -7.3052849489 ^a	-7.3054688132 -7.3054688131 ^a	-7.2290628564

^aUsing 800-term wave functions.^bReference 46.

$$\Psi = \sum_{i=1}^{N_b} \sum_{l_1=0}^2 A_i [Y_{LM}^{l_1, l_2}(\mathbf{r}_1, \mathbf{r}_2) \chi(r_1, \alpha_i, \omega) \chi(r_2, \beta_i, \omega) \chi(r_{12}, \gamma_i, \omega) + S_{pn} Y_{LM}^{l_2, l_1}(\mathbf{r}_1, \mathbf{r}_2) \chi(r_1, \beta_i, \omega) \chi(r_2, \alpha_i, \omega) \chi(r_{21}, \gamma_i, \omega)], \quad (4)$$

$$\chi(r, a, \omega) = \exp(-a\omega r), \quad (5)$$

and the bipolar harmonics $Y_{LM}^{l_1, l_2}(\mathbf{r}_1, \mathbf{r}_2)$ can be written as

$$Y_{LM}^{l_1, l_2}(\mathbf{r}_1, \mathbf{r}_2) = r_1^{l_1} r_2^{l_2} \sum_{m_1, m_2} \langle l_1 l_2 m_1 m_2 | LM \rangle Y_{l_1 m_1}(\hat{r}_1) Y_{l_2 m_2}(\hat{r}_2), \quad (6)$$

where $Y_{l m_i}(\hat{r}_j)$ denotes the usual spherical harmonics, $S_{pn} = +1$ for singlet states, $S_{pn} = -1$ for triplet states, ω is a scaling factor, N_b is the number of basis terms, and the non-linear vibrational parameters α_i, β_i , and γ_i are determined using a quasi-random process.

III. RESULTS

We have mentioned earlier that we intend, in this work, to calculate the doubly-excited 1^3D^e resonance states of two-electron positive ion Li^+ and Be^{2+} by employing the stabilization method. According to computational procedure prescribed in the previous articles on the stabilization method,¹ in the present study, we compute the energy levels $E(\omega, \mu)$ of the proposed systems by diagonalizing the atomic Hamiltonian (1) using the exponential correlated wave functions (4) in which exponent is generated by a quasi-random process. We have selected the optimized values of the

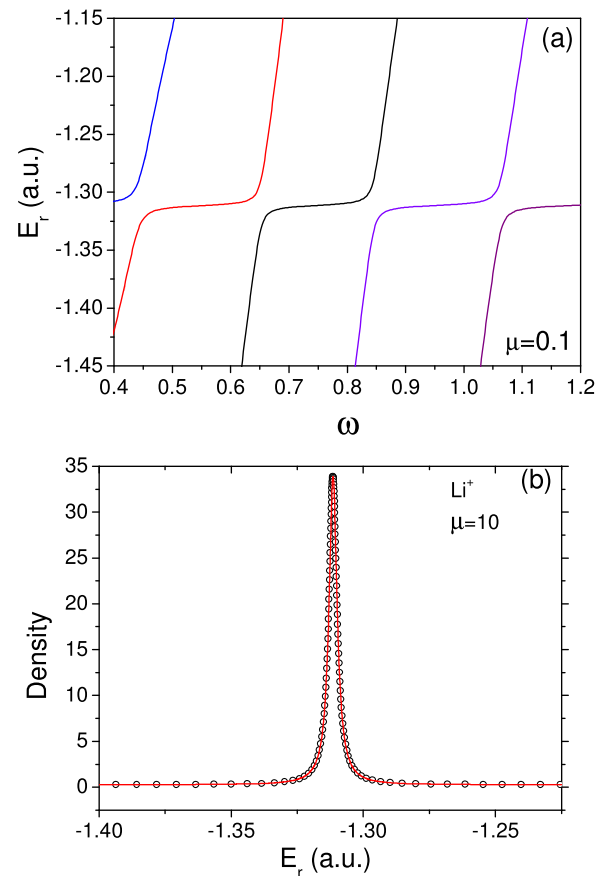


FIG. 1. (a) The Stabilization diagram for the $2p^2 \ ^1D^e$ state of Li^+ for $\mu = 0.1$. (b) The best fittings (solid line) of the density of states (circles) corresponding to the $2p^2 \ ^1D^e$ state of Li^+ for $\mu = 0.1$.

TABLE III. The $^1D^e$ resonance states of Be^{2+} for different values of the screening parameter, along with the Be^{3+} ($2s\ ^2S$) threshold energies.

$1/\mu$	$2p^2\ ^1D^e$		$2p3p\ ^1D^e$		$2s3d\ ^1D^e$		$2p4p\ ^1D^e$		$2s4d\ ^1D^e$		$Li^{2+}(2s\ ^2S)$
	E_r	$\Gamma(\times 10^{-3})$	E_r	$\Gamma(\times 10^{-3})$	E_r	$\Gamma(\times 10^{-4})$	E_r	$\Gamma(\times 10^{-4})$	E_r	$\Gamma(\times 10^{-5})$	
∞	-1.77058	3.8761	-1.37411	1.1034	-1.34617	1.611	-1.26170	4.670	-1.24941	8.93	-1.1250000000
50	-1.67229	3.867	-1.27711	1.098	-1.24937	1.56	-1.16647	4.60	-1.15454	7.99	-1.0661818501
20	-1.53109	3.82	-1.14221	1.074	-1.11538	1.33	-1.03965	4.12	-1.02920	5.49	-0.9822274203
10	-1.31149	3.66	-0.94301	0.947	-0.91907	0.773	-0.86425	2.28	-0.85847	3.42	-0.8529472079
6	-1.04729	3.34	-0.72241	0.662	-0.70583	0.285					-0.6994889358
5	-0.92675	3.13	-0.62999	0.639							-0.6302120452
4	-0.75978	2.77									-0.5349233591
3	-0.51680	2.18									-0.3973939799

TABLE IV. The $^3D^e$ resonance states of Li^+ and Be^{2+} for different values of the screening parameter, along with the Li^{2+} ($2s\ ^2S$) and Be^{3+} ($2s\ ^2S$) threshold energies.

1/ μ	2p3p $^3D^e$		2s3d $^3D^e$		2p4p $^3D^e$	2s4d $^3D^e$		Li $^{2+}$ (2s 2S) threshold energy
	Li $^+$							
	E_r	$\Gamma(\times 10^{-8})$	E_r	$\Gamma(\times 10^{-5})$	E_r	E_r	$\Gamma(\times 10^{-5})$	
∞	-1.40557	1.57	-1.36295	2.116	-1.27220	-1.25638	1.010	-1.1250000000
200	-1.38075	1.56	-1.33814	2.115	-1.24750	-1.23169	1.006	-1.1100747104
50	-1.30840	1.41	-1.26592	2.10	-1.17677	-1.16121	0.956	-1.0661818501
20	-1.17264	0.837	-1.13084	1.98	-1.04895	-1.03471	0.769	-0.9822274203
10	-0.97058	0.092	-0.93136	1.54	-0.87040	-0.86125	0.504	-0.8529472079
5	-0.64613							-0.6302120452
Be $^+$								
∞	-2.58834		-2.52706	2.893	-2.31523	-2.29191	1.9963	-2.0000000000
100	-2.51910		-2.45783	2.889	-2.24646	-2.22318	1.993	-1.9602982686
50	-2.45134		-2.39012	2.87	-2.18006	-2.15690	1.985	-1.9211862935
20	-2.25661		-2.19567	2.78	-1.99397	-1.97158	1.92	-1.8072922342
15	-2.15372		-2.09304	2.70	-1.89841	-1.87677	1.84	-1.7461820806
10	-1.95860		-1.89878	2.48				-1.6284144496
5	-1.45141		-1.39814	1.49				-1.3083391327
4	-1.23795		-1.19232	1.04				-1.1655733552

TABLE V. Comparison of $^{1,3}D^e$ resonance parameters for Li^+ and Be^{2+} for the unscreened case. The numbers in the square brackets denote the power of ten.

States	Li^+				Be^{2+}			
	Present work		Ho and Bhatia ⁴⁰		Present work		Ho and Bhatia ⁴⁰	
	E_r	Γ	E_r	Γ	E_r	Γ	E_r	Γ
$2p^2\ ^1D^e$	-1.77058	3.8761[-3]	-1.77060625	3.853[-3]	-3.33738	4.8533[-3]	-3.3374335	4.844[-3]
$2p3p\ ^1D^e$	-1.37411	1.1034[-3]	-1.3741005	1.104[-3]	-2.54028	1.4370[-3]	-2.5402875	1.426[-3]
$2s3d\ ^1D^e$	-1.34617	1.611[-4]	-1.3461705	1.57[-4]	-2.49465	3.421[-4]	-2.4946505	3.39[-4]
$2p4p\ ^1D^e$	-1.26170	4.670[-4]			-2.29966	5.534[-4]		
$2s4d\ ^1D^e$	-1.24941	8.93[-5]			-2.27914	1.885[-4]		
$2p4f\ ^1D^e$	-1.24578				-2.27492	3.02[-6]		
$2p5p\ ^1D^e$	-1.21112	2.332[-4]			-2.1897			
$2s5d\ ^1D^e$	-1.20470	4.77[-5]			-2.1787			
$2p3p\ ^3D^e$	-1.40557	1.57[-8]	-1.40556835	1.66[-8]	-2.58834		-2.58833755	9.6[-9]
$2s3d\ ^3D^e$	-1.36295	2.116[-5]	-1.362954	1.98[-5]	-2.52706	2.893[-5]	-2.527061	2.83[-5]
$2p4p\ ^3D^e$	-1.27220				-2.31523			
$2s4d\ ^3D^e$	-1.25638	1.010[-5]			-2.29191	1.9963[-5]		

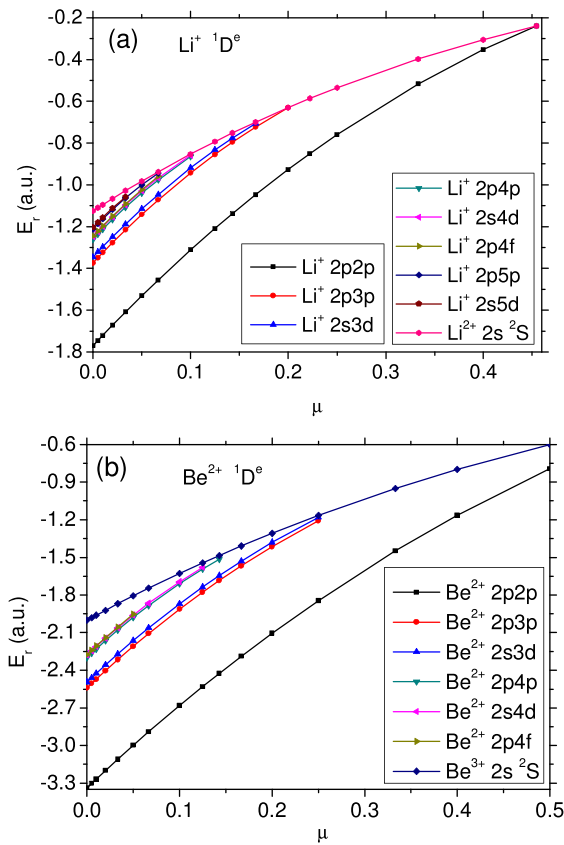


FIG. 2. The $1D^\circ$ [in (a) and (b)] resonance energy of two-electron positive ions Li^+ and Be^{2+} as a function of the Debye screening parameter μ .

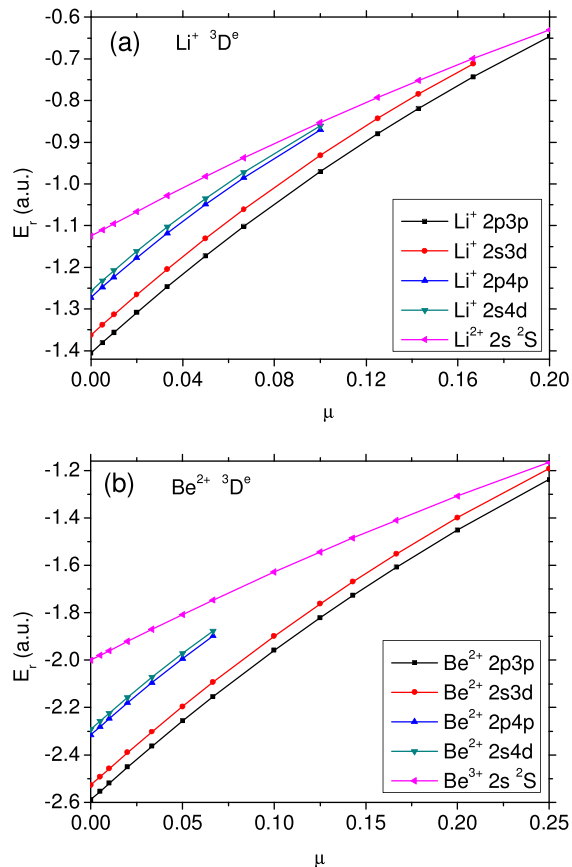


FIG. 3. The $3D^\circ$ [in (a) and (b)] resonance energy of Li^+ and Be^{2+} as a function of μ .

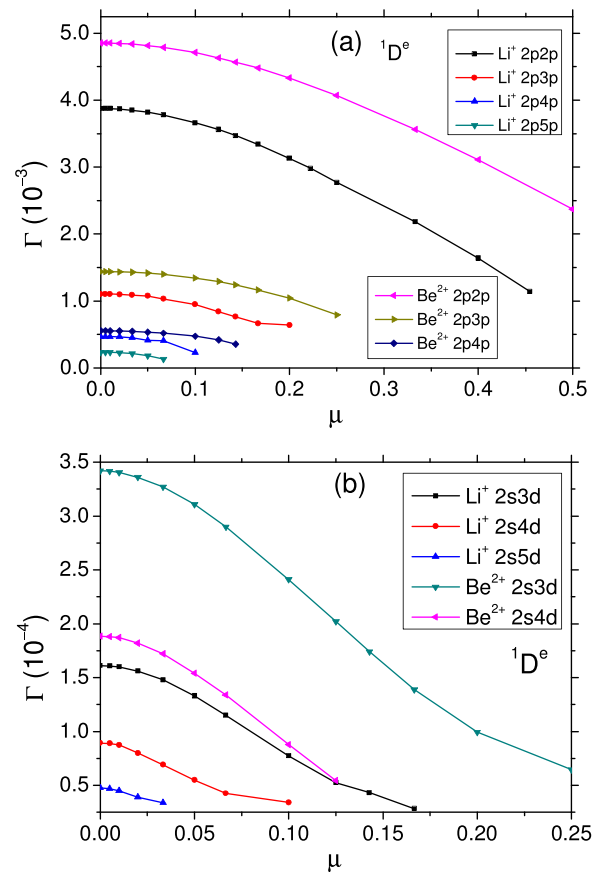


FIG. 4. The $2pnp$ $1D^\circ$ [in (a)] and $2snd$ $1D^\circ$ [in (b)] resonance widths of two-electron positive ions Li^+ and Be^{2+} below the $N=2$ thresholds of the respective subsystems as a function of the Debye screening parameter μ .

tables for $1,3D^\circ$ resonance states are obtained using 500-term wave functions. It is important to mention here that we are unable to extract resonance widths for the $2p4f$ $1D^\circ$ and $2p4p$ $3D^\circ$ states of Li^+ , and the $2p5p$ $1D^\circ$ and $2s5d$ $1D^\circ$, $2pnp$ ($n=4, 5$) $3D^\circ$ and $2p4p$ $3D^\circ$ states of Be^{2+} using 500-term wave functions. Behavior of the bound $1s3d$ $1,3D$ state energies of Li^+ and Be^{2+} as functions of the screening parameters μ are depicted in Fig. 7, along with the $1s$ $2S$ thresholds of the respective subsystems. According to the method and wave functions employed, it appears that the results obtained from the present calculations are reasonably accurate.

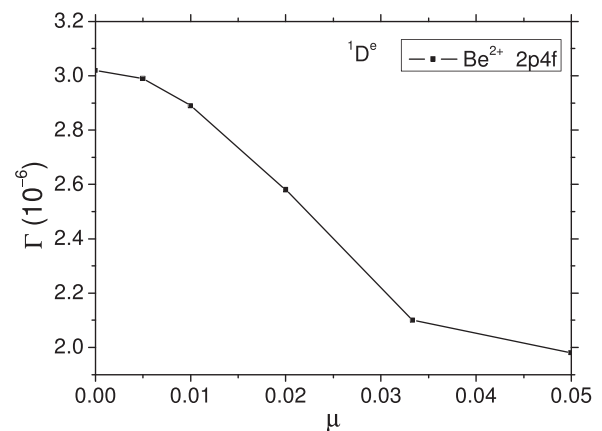


FIG. 5. The $2p4f$ $1D^\circ$ resonance width of Be^{2+} below the $N=2$ threshold of Be^{3+} as a function of the Debye screening parameter μ .

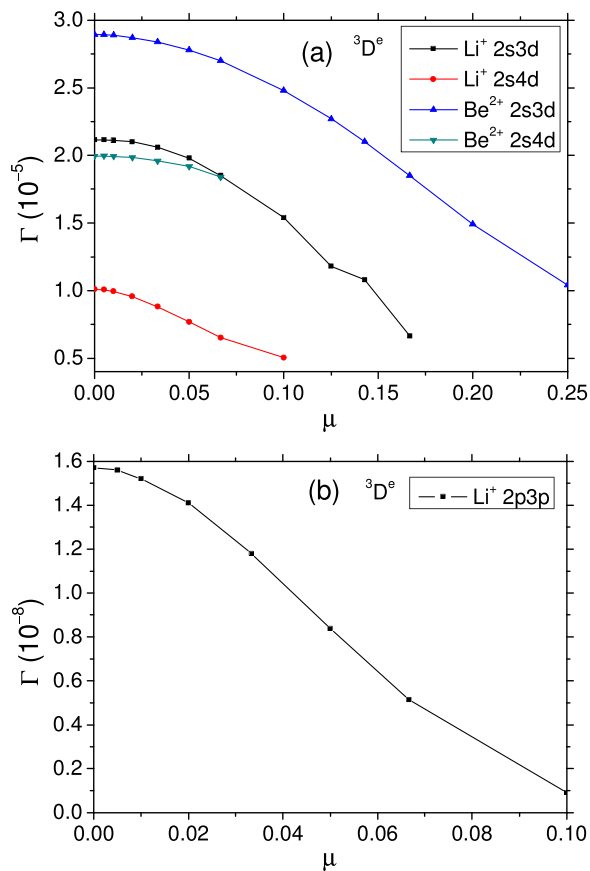


FIG. 6. The $2snd\ ^3D^\circ$ [in (a)] and $2p3p\ ^3D^\circ$ [in (b)] states resonance widths of Li^+ and Be^{2+} below the $N=2$ thresholds of the respective subsystems as functions of the Debye screening parameter μ .

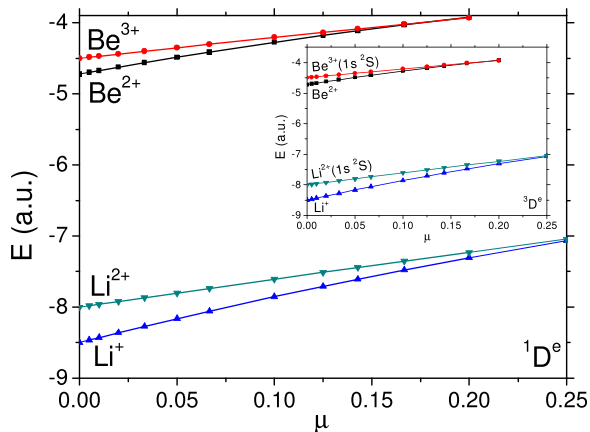


FIG. 7. The bound $1s3d\ ^{1,3}D^\circ$ states of Li^+ and Be^{2+} as functions of screening parameter along with the hydrogenic threshold energies of the respective subsystems.

IV. CONCLUSIONS

In this paper, we have presented the doubly-excited $^{1,3}D^\circ$ resonance states of two-electron positive ions ($Z=3-4$) in Debye plasmas using highly correlated exponential wave functions. We have used the stabilization method to extract resonance position and width for the $^{1,3}D^\circ$ states of Li^+ and Be^{2+} . The bound $1s3d\ ^{1,3}D$ state energies are also presented

as functions of the screened parameter. For the unscreened case, resonance parameters for $2pnp\ (n=4,5)\ ^{1,3}D^\circ$, $2snd\ (n=4,5)\ ^{1,3}D^\circ$, and $2p4f\ ^1D^\circ$ states are reported for the first time in the literature. For the screened case, our results are also reported for the first time. We hope our findings will provide useful information for future studies on this topic.

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